

Regression Analysis 2

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1. Introduction

1.1. Project Libraries and Installation

```
# Project libraries
libs <- c("tidyverse", "readxl", "knitr")

# Install only the missing ones
missing <- setdiff(libs, rownames(installed.packages()))
if (length(missing) > 0) {
  install.packages(missing, repos = "https://cloud.r-project.org")
}

# Load all of them
invisible(lapply(libs, library, character.only = TRUE))
```

2. Problem 2: Seed Germination (Logistic Regression)

In a study on seed germination, two seed types and two rootstock types were used. The goal is to determine whether the rootstock and the seed type determine whether a plant germinates. The results are in `germ.xlsx`.

2.1. Initial Description of the Data

```
germ_raw <- read_excel("data/germ.xlsx")
str(germ_raw)
```

```
tibble [831 × 3] (S3: tbl_df/tbl/data.frame)
 $ Portain   : chr [1:831] "Bean" "Bean" "Bean" "Bean" ...
 $ Semilla   : chr [1:831] "0A75" "0A75" "0A75" "0A75" ...
 $ Resultado: chr [1:831] "Germ" "Germ" "Germ" "Germ" ...
```

```
colSums(is.na(germ_raw))
```

Portain	Semilla	Resultado
0	0	0

The dataset has 831 observations (one per seed) and no missing values. All three variables are categorical:

- **Portain** (rootstock): Bean or Cucumber.
- **Semilla** (seed type): 0A73 or 0A75.
- **Resultado** (result): Germ (germinated) or NotGerm (did not germinate). This is the **response variable**.

For the model, the response is coded as 1 = germinated and 0 = did not germinate, and the explanatory variables are converted to factors. The reference levels are **Bean** and **0A73** (alphabetical order).

```
germ <- germ_raw |>
  mutate(
    germinated = if_else(Resultado == "Germ", 1, 0),
    Portain     = factor(Portain),
    Semilla     = factor(Semilla)
  )
```

2.1.1. Frequency Tables

```
germ |> count(Portain) |> kable(caption = "Seeds per rootstock")
```

Portain	n
Bean	395
Cucumber	436

```
germ |> count(Semilla) |> kable(caption = "Seeds per seed type")
```

Semilla	n
OA73	264
OA75	567

```
germ |> count(Resultado) |> kable(caption = "Germination result")
```

Resultado	n
Germ	424
NotGerm	407

2.1.2. Germination Proportion by Group

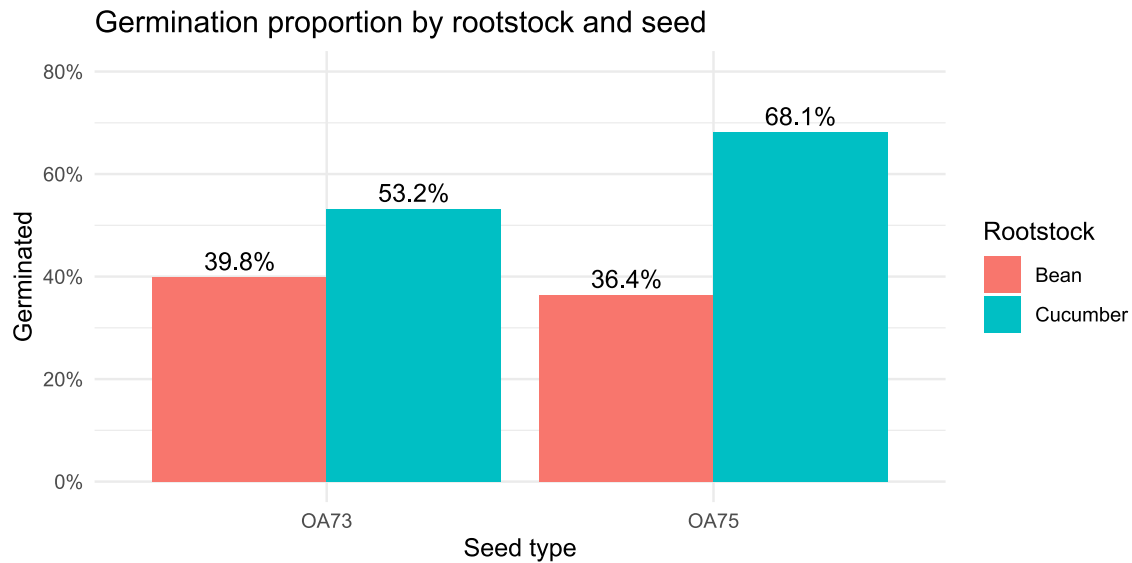
```
germ_summary <- germ |>
  group_by(Portain, Semilla) |>
  summarise(
    n = n(),
    n_germinated = sum(germinated),
    proportion = mean(germinated),
    .groups = "drop"
  )

germ_summary |> kable(digits = 3, caption = "Germination by rootstock and seed")
```

Portain	Semilla	n	n_germinated	proportion
Bean	OA73	123	49	0.398
Bean	OA75	272	99	0.364
Cucumber	OA73	141	75	0.532
Cucumber	OA75	295	201	0.681

```
germ_summary |>
  ggplot(aes(x = Semilla, y = proportion, fill = Portain)) +
```

```
geom_col(position = "dodge") +
  geom_text(aes(label = scales::percent(proportion, accuracy = 0.1)),
            position = position_dodge(width = 0.9), vjust = -0.4) +
  scale_y_continuous(labels = scales::percent, limits = c(0, 0.8)) +
  labs(title = "Germination proportion by rootstock and seed",
       x = "Seed type", y = "Germinated", fill = "Rootstock") +
  theme_minimal()
```



Observations:

- Seeds on the **Cucumber** rootstock germinate more often than those on **Bean**, for both seed types.
- The effect of the seed type depends on the rootstock: with **Bean**, OA75 germinates slightly *less* than OA73; with **Cucumber**, OA75 germinates clearly *more*. This suggests an **interaction** between rootstock and seed, which is tested in the model.

2.2. Logistic Regression Model

2.2.1. Main Effects Model

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{Cucumber} + \beta_2 \text{OA75}$$

```
logit_main <- glm(germinated ~ Portain + Semilla, family = binomial, data = germ)
summary(logit_main)
```

Call:

```
glm(formula = germinated ~ Portain + Semilla, family = binomial,
    data = germ)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.7005	0.1507	-4.648	3.36e-06 ***
PortainCucumber	1.0647	0.1442	7.383	1.55e-13 ***
SemillaOA75	0.2705	0.1547	1.748	0.0804 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1151.7 on 830 degrees of freedom
 Residual deviance: 1092.6 on 828 degrees of freedom
 AIC: 1098.6

Number of Fisher Scoring iterations: 4

2.2.2. Model with Interaction

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \text{Cucumber} + \beta_2 \text{OA75} + \beta_3 (\text{Cucumber} \times \text{OA75})$$

```
logit_int <- glm(germinated ~ Portain * Semilla, family = binomial, data = germ)
summary(logit_int)
```

Call:

```
glm(formula = germinated ~ Portain * Semilla, family = binomial,
    data = germ)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.4122	0.1842	-2.238	0.0252 *
PortainCucumber	0.5401	0.2498	2.162	0.0306 *
SemillaOA75	-0.1459	0.2232	-0.654	0.5132
PortainCucumber:SemillaOA75	0.7781	0.3064	2.539	0.0111 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 1151.7 on 830 degrees of freedom
Residual deviance: 1086.2 on 827 degrees of freedom
AIC: 1094.2
```

```
Number of Fisher Scoring iterations: 4
```

2.2.3. Model Comparison (Likelihood Ratio Test)

- H_0 : the interaction is not needed ($\beta_3 = 0$).
- H_1 : the interaction is needed ($\beta_3 \neq 0$).

```
anova(logit_main, logit_int, test = "Chisq")
```

Analysis of Deviance Table

```
Model 1: germinated ~ Portain + Semilla
```

```
Model 2: germinated ~ Portain * Semilla
```

```
Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      828      1092.6
2      827      1086.2  1    6.4081  0.01136 *
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Observations: the p-value of the likelihood ratio test is less than 0.05, so H_0 is rejected: the interaction model fits significantly better, and it also has a lower AIC (1094.2 vs 1098.6). The interaction model is the one interpreted below.

2.3. Interpretation of the Coefficients

In logistic regression the coefficients are on the **log-odds** scale. Exponentiating them gives **odds ratios (OR)**, which are easier to interpret.

```
odds_ratios <- exp(cbind(OR = coef(logit_int), confint.default(logit_int)))
odds_ratios |> kable(digits = 3, caption = "Odds ratios with 95% confidence intervals")
```

	OR	2.5 %	97.5 %
(Intercept)	0.662	0.462	0.950
PortainCucumber	1.716	1.052	2.800
SemillaOA75	0.864	0.558	1.338
PortainCucumber:SemillaOA75	2.177	1.194	3.970

```
b <- coef(logit_int)
```

- **Intercept** ($\beta_0 = -0.4122$): for the reference group (Bean + OA73), the odds of germinating are $e^{\beta_0} = 0.662$. That is, a germination probability of 0.398.
- **PortainCucumber** ($\beta_1 = 0.5401$, $p < 0.05$): with seed OA73, using the Cucumber rootstock multiplies the odds of germinating by 1.716 compared with Bean (a 71.6% increase).
- **SemillaOA75** ($\beta_2 = -0.1459$, $p > 0.05$): with the Bean rootstock, seed OA75 has odds of germinating 0.864 times those of OA73. This effect is **not significant**: on Bean, the seed type does not change germination.
- **Interaction** ($\beta_3 = 0.7781$, $p < 0.05$): the effect of the Cucumber rootstock is 2.177 times larger for seed OA75 than for OA73. Combining the coefficients:
 - With OA75, Cucumber vs Bean: $OR = e^{\beta_1 + \beta_3} = 3.737$.
 - With Cucumber, OA75 vs OA73: $OR = e^{\beta_2 + \beta_3} = 1.882$.

2.3.1. Predicted Probabilities

```
germ_summary |>
  select(Portain, Semilla) |>
  mutate(predicted = predict(logit_int, newdata = pick(everything()), type =
    "response")) |>
  kable(digits = 3, caption = "Predicted germination probability")
```

Portain	Semilla	predicted
Bean	OA73	0.398
Bean	OA75	0.364
Cucumber	OA73	0.532
Cucumber	OA75	0.681

Conclusion: both the rootstock and the seed type determine germination, but not independently. The highest germination probability (0.681) is obtained with the combination Cucumber + OA75. On Bean, the seed type makes no significant difference.